forward biased. Since electrical energy is converted into light energy, this process is also called electroluminescence. The circuit symbol of LED is shown in Figure 10.22(a). The direction of arrows indicates that light is emitted from the diode.

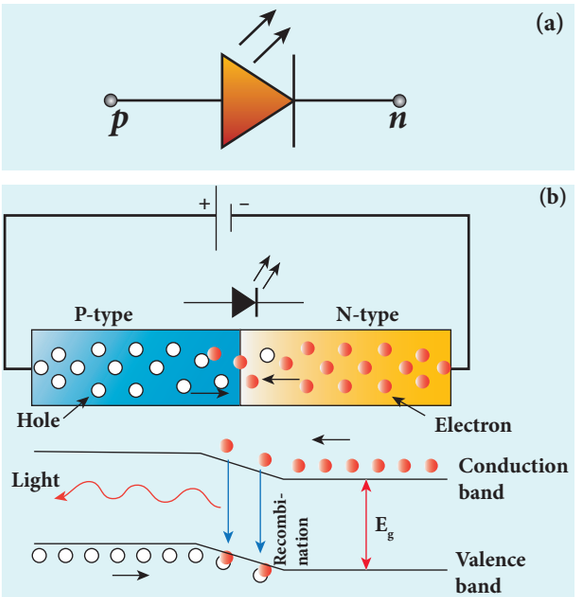


Figure 10.22 (a) Circuit symbol of LED (b) Schematic diagram to explain recombination process

When the *p*-*n* junction is forward biased, the conduction band electrons on *n*-side and valence band holes on *p*-side diffuse across the junction. When they cross the junction, they become excess minority carriers (electrons in *p*-side and holes in *n*-side). These excess minority carriers recombine with oppositely charged majority carriers in the respective regions, i.e. **the electrons in the conduction band recombine with holes in the valence band** as shown in the Figure 10.22(b).

During recombination process, energy is released in the form of light (radiative) or heat (non-radiative). For radiative recombination, a photon of energy $h\nu$ is emitted. For non-radiative recombination, energy is liberated in the form of heat.

The colour of the light is determined by the energy band gap of the material. Therefore, LEDs are available in a wide range of colours such as blue (SiC), green (AlGaP) and red (GaAsP). Now a days, LED which emits white light (GaInN) is also available.

Applications

The light emitting diodes are used in

- indicator lamps on the front panel of the scientific and laboratory equipments.
- seven-segment displays.
- traffic signals, emergency vehicle lighting etc.
- remote control of television, air-conditioner etc.

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EXAMPLE 10.4

Determine the wavelength of light emitted from LED which is made up of GaAsP semiconductor whose forbidden energy gap is 1.875 eV. Mention the colour of the light emitted (Take $h = 6.6 \times 10^{-34}$ Js).

Solution

$$E_g = \frac{hc}{\lambda}$$

Therefore,

$$\lambda = \frac{hc}{E_g} = \frac{6.6\times10^{-34}\times3\times10^{8}}{1.875\times1.6\times10^{-19}}$$
$$= 660 \text{ nm}$$

The wavelength 660 nm corresponds to red colour light.

ii) Photodiodes

A *p*-*n* junction diode which converts an optical signal into electric signal is known as photodiode. Thus, the operation

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